

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I am **B.Chaithanya Prakash Reddy(192425255)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks
Understanding of Concepts	25%			
Explanation	25%			
Application	20%			
Organization	15%			
Accuracy	10%			
Clarity	5%			

Total Marks /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and wellstructured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Question 1

The missing layers in the given OSI model concept map are Session Layer and Data Link Layer.

- The Session Layer (Layer5) is responsible in establishing, managing, synchronizing, and terminating communication sessions between two devices. It ensures that communication remains organized throughout the data exchange process. If the connection is interrupted, the Session Layer can

resume communication from synchronization points instead of restarting the entire transmission. This improves the reliability and efficiency of network communication.

- The Data Link Layer (Layer 2) is responsible for converting packets into frames and ensuring reliable communication between devices connected to the same local network. It uses MAC (Media Access Control) addresses for local device identification and performs error detection using techniques such as Cyclic Redundancy Check (CRC). If an error is detected during transmission, the damaged frame can be retransmitted, reducing the chances of data corruption.
- Both layers play an important role in reliable communication. The Session Layer maintains stable communication sessions, while the Data Link Layer ensures accurate and error-free transmission of data between neighbouring devices. Together, these layers contribute to efficient, secure, and reliable communication in the OSI model.

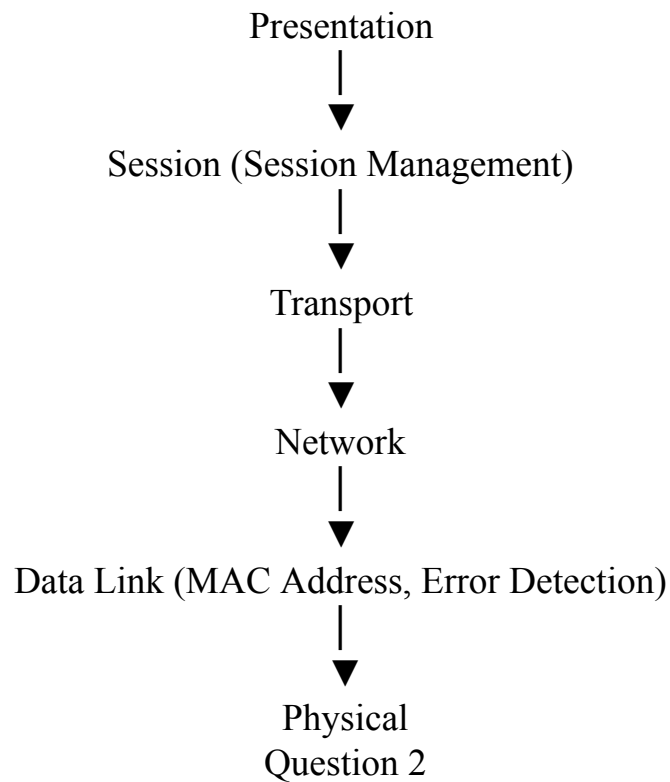
Conclusion

- The Session Layer manages communication sessions, and the Data Link Layer provides reliable local data transfer with framing, MAC addressing, and error detection. These functions help achieve dependable communication across the network.

Question 2 :

The completed OSI sequence is:

Application
↓
▼



- In the OSI (Open Systems Interconnection) model, data is transformed into different Protocol Data Units (PDUs) as it moves through the network layers. This process is called encapsulation. Each layer adds its own header (and sometimes a trailer) to the data, allowing it to perform specific networking functions. At the receiver's end, the reverse process, called decapsulation, removes these headers layer by layer to retrieve the original data.
- The Transport Layer receives data from the upper layers and divides it into Segments. Each segment contains TCP or UDP information that provides sequencing, error recovery, flow control, and reliable end-to-end communication.
- The Network Layer encapsulates each segment into a Packet by adding an IP header. The packet contains the source and destination IP addresses, enabling routers to determine the best path for delivering data across different networks.
- The Data Link Layer converts each packet into a Frame by adding a MAC header and trailer. The frame contains source and destination MAC addresses and performs error detection using techniques such as Cyclic Redundancy Check (CRC). This ensures reliable communication within the local network.
- The Physical Layer converts the frame into Bits (0s and 1s). These bits are transmitted as electrical signals through cables, optical signals through fiber optics, or radio waves in wireless communication.
- This transformation of data units enables each OSI layer to perform its specific responsibilities efficiently. By adding addressing information, error

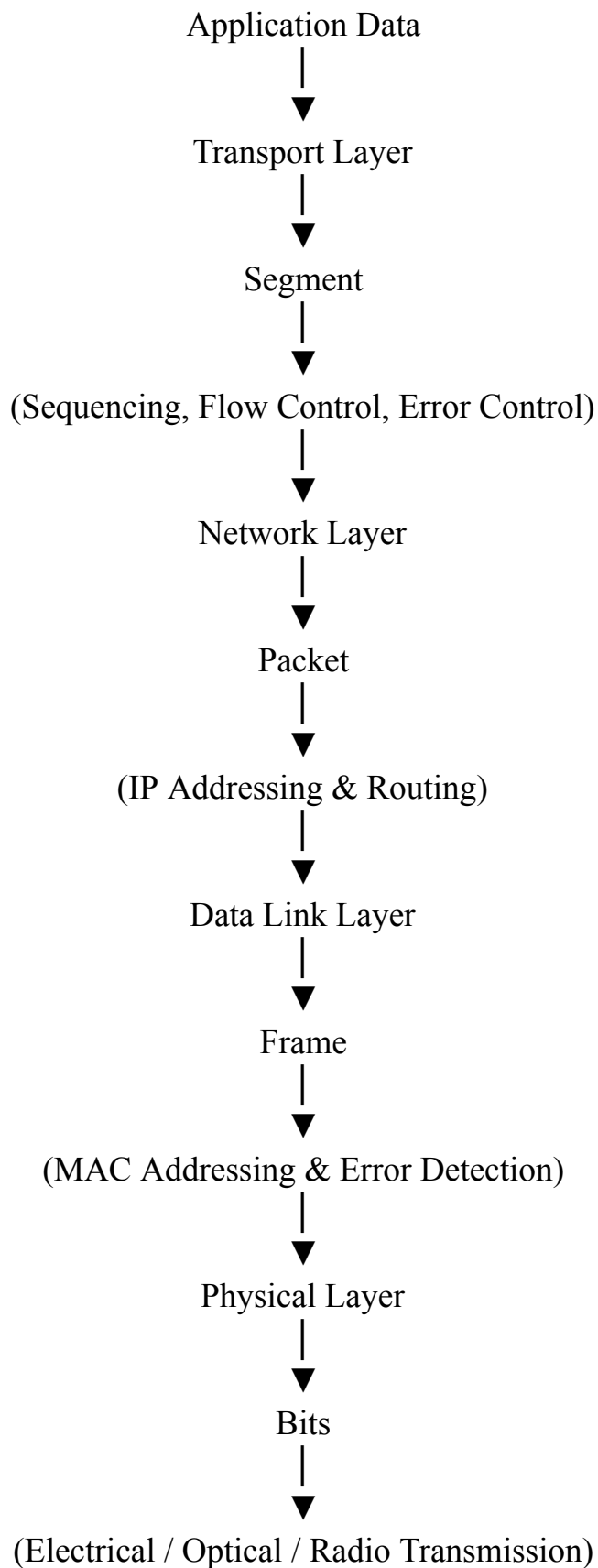
detection mechanisms, and transmission details, the OSI model ensures that data reaches the correct destination accurately and reliably.

OSI Layer	Protocol Data Unit (PDU)	Purpose
Transport Layer	Segment	Ensures reliable end-to-end communication through sequencing, acknowledgments, and flow control.
Network Layer	Packet	Provides logical addressing (IP Address) and routing between networks.
Data Link Layer	Frame	Provides physical addressing (MAC Address) and error detection.
Physical Layer	Bits	Transmits binary data over the communication medium.

- The transformation of data into different units is essential because each OSI layer has a unique role:
- Segments organize the data for reliable transport and ensure that lost or damaged data can be retransmitted.
- Packets carry IP addressing information, allowing routers to forward data to the correct destination across multiple networks.
- Frames use MAC addresses to deliver data within the local network and detect transmission errors.
- Bits represent the binary form of data that can be transmitted over wired or wireless communication channels.
- Without this step-by-step transformation, communication between computers would not be reliable, organized, or efficient.

Conclusion

The process of transforming Data → Segment → Packet → Frame → Bits is known as encapsulation. It allows each OSI layer to add the necessary information required for communication, ensuring accurate addressing, routing, error detection, and reliable transmission. At the receiving end, decapsulation reconstructs the original data, enabling successful communication between devices.



Question 3

- In the OSI model, the Data Link Layer and the Network Layer work together to ensure that data is delivered successfully from the sender to the receiver. The Data Link Layer uses MAC (Media Access Control) addresses for communication within the same local network (LAN), while the Network Layer uses IP (Internet Protocol) addresses for communication between different networks.

- The Data Link Layer is responsible for local delivery. Every network device has a unique MAC address assigned to its network interface card (NIC). When a frame is transmitted within the same LAN, switches use the destination MAC address to forward the frame to the correct device. This allows accurate communication between devices connected to the same local network.
- The Network Layer is responsible for global delivery. Each device connected to a network has a unique IP address. Routers use the source and destination IP addresses to determine the best path for forwarding packets across multiple interconnected networks until they reach the destination network.
- When data is transmitted from one computer to another over the Internet, both MAC and IP addresses are required. The IP address identifies the destination network, while the MAC address identifies the specific device within that network. As the packet moves through routers, the IP address remains the same, but the MAC address changes at each hop because every local network uses different physical addresses.
- Thus, the combination of MAC addressing for local delivery and IP addressing for global delivery ensures that data reaches the correct destination accurately and efficiently.

Analysis:

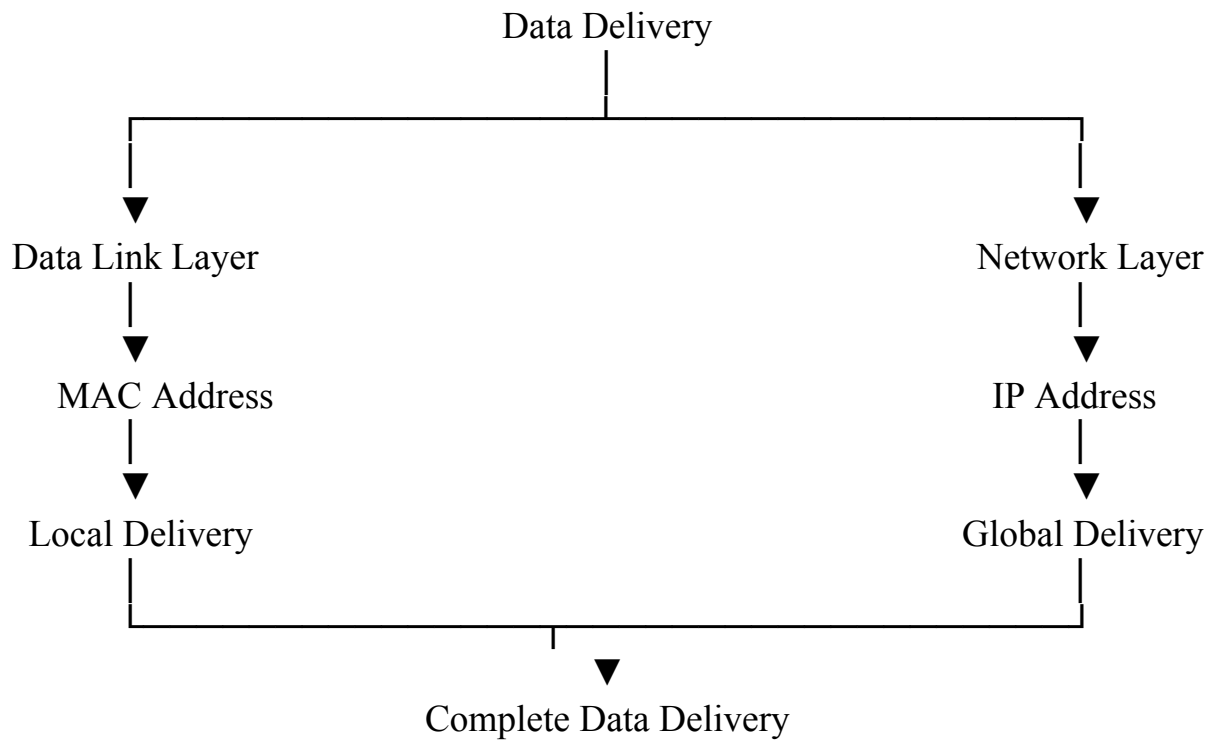
The relationship between the Data Link Layer and the Network Layer is essential for complete communication.

- The Data Link Layer ensures that frames are delivered correctly within a local network using MAC addresses.
- The Network Layer ensures that packets are routed across different networks using IP addresses.
- Routers read IP addresses to determine the destination network, while switches read MAC addresses to deliver data to the correct device within that network.
- Without the Data Link Layer, local communication would fail.
- Without the Network Layer, communication between different networks would not be possible.

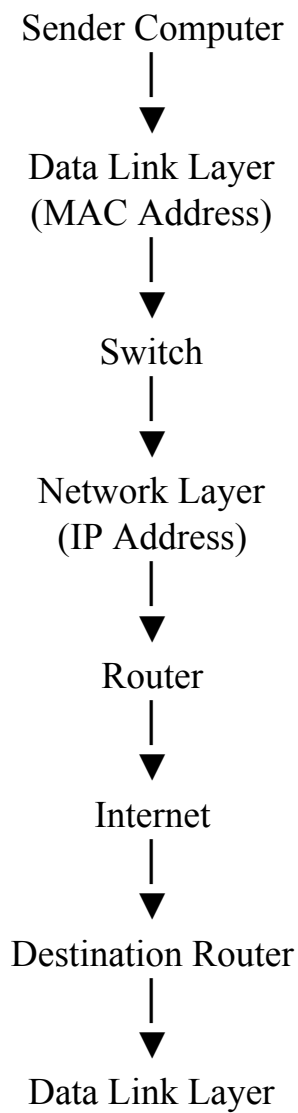
Therefore, both layers work together to provide reliable end-to-end data delivery.

Conclusion :

The Data Link Layer uses MAC addresses to provide reliable local delivery, while the Network Layer uses IP addresses to provide global delivery across multiple networks. Their combined operation ensures that data travels from the source device to the destination device successfully, making communication over both local networks and the Internet possible.



Communication Flow:



(MAC Address)



Destination Computer